

# Sachan Rai

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## Experience

### Software Engineering Intern

Jan 2026 - Apr 2026

#### Narrative Projects

Toronto, ON

- Designed an **inbound email parsing webhook**, routing mail to the service via an **MX record** and processing **100+** daily messages, recovering threads where the comms inbox was added later via **header and subject matching**
- Replaced an in-app **text-style messaging system** with an email-driven Comms Hub in **Node.js + TypeScript** on **Cloud Run**, AI-tagging emails to products at **90% accuracy** across **1000+ brand-owner products**
- Self-hosted the **OpenClaw AI framework** on a **GCP Compute Engine VM** to power email-driven order intake and daily inbox digests, scoped under a **dedicated service account** with **IAM-controlled SSH**

### System Operations Intern

May 2025 - Aug 2025

#### Canadian Analytical Laboratories

Mississauga, ON

- Audited **200+ IT assets** across **4 departments** by building a standardized **ITAM tracker in Excel**, surfacing end-of-life devices and warranty coverage gaps, informing the infrastructure refresh and budget planning cycle
- Configured workstations for **lab, regulatory, QA, and admin teams** by deploying standardized Windows builds and installing **chromatography systems (Waters Empower, Agilent OpenLab)** under **GMP/Health Canada standards**
- Coordinated rollouts with scientists, IT leads, and regulatory staff by scheduling cutovers around active assay runs, delivering hardware and software upgrades with **zero downtime** across ongoing **clinical trial documentation**

### Controls Engineer

Sep 2024 - Apr 2025

#### Waterloo Rocketry

Waterloo, ON

- Engineered a **mechanical airbrake system** in **SOLIDWORKS**, designing symmetric deployable drag panels that fit within the rocket's airframe and mass budget for a projected **20% improvement in aerodynamic efficiency**
- Designed and tested a **custom PCB airbrake control board** in **KiCAD**, integrating sensor inputs and **filtering noisy telemetry signals** in real time to deliver deployment commands during high-speed boost and coast phases
- Partnered with the **avionics and recovery subteam** to integrate the airbrake PCB with the main flight computer's data and power buses, resolving **cross-subsystem integration issues** to lock in a flight-ready revision

## Projects

### 🔗 Apex - AI Racing Telemetry Coach | Python, FastAPI, React, Recharts, PostgreSQL, NumPy, UDP/Sockets, Anthropic API

- Built a **real-time ingestion layer** that parses **F1 25's binary UDP telemetry** at up to **60Hz** with **Python's struct module**, segmenting a continuous stream into discrete laps via **lap-number detection** and feeding the analysis engine through a single-source-of-truth codec round-trip **validated over UDP** to **±1 km/h** and **±0.5m**
- Engineered a **NumPy** lap-analysis engine that resamples telemetry onto a common distance grid and reconstructs **cumulative lap time** from speed to localize where a lap loses time against a reference, with **track-agnostic corner detection** placing apexes within **±50m** of ground truth using **steering-confirmed speed minima**, all backed by a **23-test suite**

### 🔗 Atlas AI - PCB Defect Detection | PyTorch, React, FastAPI, PostgreSQL, Redis, Docker, AWS

- Trained a **PyTorch CNN** on a dataset of over **1,300 PCB images** achieving **90%+ accuracy** across six defect categories, served via a **FastAPI + Redis** backend delivering **real-time inference under 120ms per image**
- Piped predictions into an **LLM microservice** that auto-generates **PDF diagnostic reports** for inspectors, summarizing detected defects with bounding boxes and severity ratings, cutting average inspection time by **75%**

### 🔗 LifeSync - AI Wellness Platform | React Native, Node.js, Python, PostgreSQL, FastAPI, AWS, Docker

- Built a **cross-platform React Native** health app with **GPT-4 RAG** over **500+ nutrition and fitness documents**, delivering context-aware meal planning and exercise coaching via **LLM-powered analysis** tailored to user goals
- Designed a **Node.js + FastAPI** microservices backend with **PostgreSQL + Redis** query optimization reducing database load by **75%** and maintaining **sub-100ms** response times, all deployed on **AWS ECS** via **Docker**

## Education

### University Of Waterloo

Waterloo, ON

Bachelor of Applied Sciences in Computer Engineering

Sep 2024 - Apr 2029

## Technical Skills

**Languages:** C, C++, Python, Java, C#, JavaScript, TypeScript, HTML, CSS, VHDL, Verilog

**Frameworks & Libraries:** PyTorch, TensorFlow, React, React Native, Node.js, FastAPI, MongoDB, PostgreSQL, Redis

**Cloud & DevOps:** AWS (ECS, S3, RDS), GCP (Cloud Run, Compute Engine), Docker, Kubernetes, SendGrid

**Developer Tools:** Git, MATLAB, KiCAD, SOLIDWORKS, VS Code